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Sensitivity and uncertainty analysis of the APSIM-wheat model: Interactions between cultivar, environmental, and management parameters

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ABSTRACT

Process-based crop models use many cultivar parameters to simulate crop growth. Usually, these parameters cannot be directly measured and need to be calibrated when the crop model is applied to a new environment or a new cultivar. Determining the relative importance of the cultivar parameters to the specific outputs could streamline the calibration of crop models for new cultivars. Sensitivity analysis can quantify the influence of model input parameters on model outputs. We applied the variance-based global sensitivity analysis to the wheat module of the Agricultural Production Systems sIMulator (APSIM) for the first time and calculated the sensitivity of four outputs including yield, biomass, flowering day, and maturity day to ten cultivar parameters including both the main and total effects sensitivity indices. We explored the effects of changing climate, soil, and management practices on parameter sensitivity by analyzing two fertilization rates (0 and 100 kg N ha^{-1}), across five sites in Australia's cereal-growing regions. Uncertainties for the four outputs with varying cultivar parameters, climate–soil conditions and management practices were evaluated. We found that yield was most sensitive to the cultivar parameters that determine the yield component (grains per gram stem, max grain size, and potential grain filling rate) and the phenology parameters that determine length of the reproductive stages (thermal time from floral initiation to flowing and thermal time from start grain filling to maturity). All ten cultivar parameters affected biomass, amongst which the parameters of vernalization sensitivity and thermal time from floral initiation to flowering were the most influential. Fertilization influenced the rank order of parameter sensitivities more strongly than climate–soil conditions for yield and biomass outputs. Under 0 kg N ha^{-1}, with the variation of cultivar parameters simulated yield varied from 64 to 3559 kg ha^{-1} (minimum and maximum), biomass from 693 to 12,864 kg ha^{-1}. Fertilization of 100 kg N ha^{-1} increased the maximum yield to 9157 kg ha^{-1} and biomass to 22,057 kg ha^{-1}. We conclude that to minimize cultivar-related uncertainty, cultivar parameters should be carefully calibrated when applying the APSIM-wheat model to a new cultivar in a new environment. By targeting the most influential phenological parameters for calibration first and then the yield component parameters, the calibration of APSIM can be streamlined.

1. Introduction

Process-based crop models are frequently used as a scientific tool to study the impacts of changes in management and environment on yields, often with a focus on addressing global challenges such as climate change, and food and energy security (Ahuja et al., 2002; Brisson et al., 2003; Bryan et al., 2010a; Luo et al., 2005b; Ma et al., 2009). Different cultivars of the same crop may behave distinctly under varying growth environments. Process-based crop models mimic these cultivar-related behaviours by using a set of cultivar parameters which adjust crop development, phenology, partitioning, and reproduction performance. However, it is often difficult to obtain accurate values for these parameters. A common practice to estimate them is fitting the simulated results to measured data (Jansen and Painter, 1974; Makowski et al., 2006; Tattersall and Sussman, 1975) using methods such as regression or

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iterative optimization with Monte Carlo sampling and simulation (Aggarwal, 1995; Lamboni et al., 2009; Larget and Simon, 1999; Martens and Martens, 2000; Román-Paoli et al., 2000; Thorburn et al., 2001; Wallach et al., 2001). The computational costs of calibration are generally in proportion to the number of parameters and the complexity of the model. When a large number of parameters are involved, an iterative optimization method such as Bayesian calibration is prohibitive for relatively slow models (e.g. those with a long run time for a single simulation). To streamline the calibration process—making it faster and computationally cheaper—non-influential parameters need to be excluded and effort targeted towards the most influential ones (Minunno et al., 2013; Nossent et al., 2011; Sau et al., 1999; White et al., 2008).

Sensitivity analysis is an approach for evaluating the response of model outputs to variation in model input parameters, for quantifying the importance of input parameters, and for exploring model structure (Saltelli et al., 2008). Sensitivity analysis can be classified into two categories—local and global—according to the strategy used to explore the parameter space. Local methods, also called *derivative-based* or *one-at-a-time* methods, involve changing one parameter at a time around a basis point while keeping the other parameters at nominal values (Cacuci, 1981; Turanyi and Rabitz, 2000). It has a low computational cost and is relatively easily implemented, but suffers several shortcomings including a heavy dependence on the base value of the input parameters, instability for non-linear models, and the inability to detect interactions between parameters (Cacuci, 2003; Saltelli et al., 2008, 2006). Global sensitivity analysis overcomes these drawbacks. Global methods explore the entire multi-dimensional parameter space simultaneously. For a specific output variable, the influence of single parameters and the interactions between parameters can be quantified (Saltelli et al., 2008). Several global sensitivity analysis methods exist. Yang (2011) evaluated five of the commonly used techniques including: variance-based (Sobol, 1990); Morris (Morris, 1991); linear regression (Helton, 1993); regionalized sensitivity analysis (Spear and Hornberger, 1980); and non-parametric smoothing (Young, 2000). The author found that the variance-based methods were robust but computationally expensive; the Morris method was effective in eliminating insignificant parameters with limited computational cost; and the non-parametric smoothing was reliable in quantifying the main effects and low-order interactions but was not reliable for non-linear models.

Sensitivity analysis has been widely used in quantifying the importance of parameters and investigating the structure of crop models, both local (Kumar et al., 2013) and global (Confalonieri, 2010; DeJonge et al., 2012; Ma et al., 2000). In a crop model, the outputs are determined by the model structure, parameters, and input data. The importance of one parameter is not only correlated with the model structure, but also the values of other parameters and input data (Song et al., 2013; Wallach et al., 2006). For example, the importance of a physiological parameter could be altered by irrigation practice, soil properties, and climate conditions (DeJonge et al., 2012; Lamboni et al., 2009). Thus, the model outputs could be sensitive to both individual parameters and combinations of them (Pogson et al., 2012). For crop model calibration at a regional scale, the combinations of climate–soil-management are numerous. The number of parameters assessed and the number of iterative runs were frequently compromised to keep the computation load and time in an acceptable range (Angulo et al., 2013; Xiong et al., 2008). An integrated assessment of the sensitivities of model outputs to the management options, climate, and soil conditions is required to provide further insights in model calibration and application (Confalonieri et al., 2010).

In this study, we aimed to understand the sensitivity of the key outputs of a widely used crop model—the Agricultural Production Systems sIMulator (APSIM)—to cultivar parameters and the influence of environmental conditions and management practices. We used the extended Fourier Amplitude Sensitivity Test (extended FAST) variance-based global sensitivity analysis method to study the sensitivity of four outputs—yield, biomass, flowering day, and maturity day—to ten cultivar parameters. The effects of fertilization rates and climate–soil conditions were evaluated by their impacts on the order of sensitivity indices. We also quantified the uncertainty in the four outputs using the outputs from the variance-based sensitivity analysis. The results were verified by interpretation against the known structure of the APSIM model. The broader implications for crop model calibration are discussed with the aim of streamlining the process of calibration of new cultivars in crop models.

Table 1
Phenological development stages included in APSIM-wheat.

Stage code	Stage name	Starting processes
1	Sowing	Seed germination
2	Germination	Emergence, leaf initiation
3	Emergence	Vegetative growth (LAI, DM), water/N uptake
4	End of Juvenile Stage	Photoperiodism
5	Floral Initiation/terminal spikelet	Spikelet initiation Rapid stem growth
6	Anthesis	Setting grain numbers
7	Start of Grain Filling	Active grain growth
8	End of Grain Filling	Maturity
9	Physiological Maturity	Grain moisture loss
10	Harvest Ripe	
11	End Crop	

Adopted from http://www.apsim.info/wiki/Wheat.ashx.

2. Methods

2.1. APSIM and the wheat-module

APSIM is a flexible framework accommodating a variety of environmental and crop modules (crop, soil, climate, management, etc.) via a plug-in mechanism (Holzworth et al., 2010, 2011; Keating et al., 2003; McCown et al., 1996). Variables of different models can be dynamically linked by an xml project file. The central engine of APSIM then interprets the project file and coordinates the communication among different modules. APSIM uses a plant model to generalize the common physiological principles of crops. A specific crop model (e.g. APSIM-wheat) can be derived from the generalized plant model by modifying the relevant parameters regulating the specific properties of the crop (Wang et al., 2002).

APSIM-wheat can simulate both spring and winter wheat cultivars. APSIM-wheat simulates wheat growth on a daily time-step (Asseng et al., 1998), driven by daily weather data with influence from soil moisture and nutrient conditions as well as management practices. It uses 11 crop stages and 10 phases (time between stages) to define the phenological development (Table 1). The start of each stage is determined by the accumulation of thermal time except for the stage from sowing to germination which is mainly determined by soil water content. The thermal time is calculated from the difference between base temperature and 3-hourly crown temperatures derived from the daily maximum and minimum temperatures. The daily thermal time values could be further impacted by water-nutrient stress, photoperiod, and vernalization depending on the development stage and cultivar. The thermal time is then accumulated to determine the phonological development of the crop. The biomass accumulation is based on radiation use efficiency (RUE). The RUE could be reduced by nitrogen stress and extreme high or low temperatures using an interpolation function. Biomass partitioning rates to different plant parts vary with crop development stage and re-translocation begins at the stage of starting grain filling.

The SoilWater module updates the values of soil water status according to the amount of soil water assimilated by the crop, soil evaporation, surface runoff, bottom drainage and irrigation. The soil water is mainly determined by three parameters, namely the lower limit (LL15), drained upper limit (DUL) and saturated (SAT) volumetric water content. The runoff on the soil surface due to intensive rainfall is calculated using the USDA-Soil Conservation Service model with different response curves that can be changed by the user. Residue cover and tillage decrease runoff. When the soil water in the bottom layer is larger than the SAT, it will run out as drainage. When the soil is sufficiently wet, the evaporation is set equal to potential evaporation rate, which is quantified by the Priestley and Taylor (1972) method. When the soil water content is lower than a threshold value, the rate of evaporation will be decreased to less than the potential evaporation. Residue cover reduces the potential soil evaporation (Ritchie, 1972) according to the coverage percentage.

The SoilN module mimics the cycling of carbon and nitrogen in the soil and updates the nutrient status that potentially limits the crop growth (Huth et al., 2010). The soil organic matter (SOM) is divided into two pools (biom and hum). The flows between the different pools are calculated based on carbon, while the nitrogen flows depending on the C:N ratio of the receiving pools, which are constant through time.

The APSIM-wheat module has been widely validated against trial data across various environments (Asseng et al., 2002; Chen et al., 2010) and applied at site, field, catchment, and continental scales (Bryan et al., 2010b; Balwinder-Singh et al., 2011; Huth et al., 2002; Luo et al., 2005a, 2007; Zhao et al., 2013b).

2.2. Study sites

Five sites across Australian cropping lands were chosen to study the effects of climate and soil condition on the sensitivity of the APSIM-wheat module (Fig. 1). The soil data for the sites were sourced from the APSoil database (Dalgliesh et al., 2012). For each site, 21-year (1990–2010) daily climate data from the nearest climate station were obtained from the SILO weather database produced by Australian Bureau of Meteorology (http://www.longpaddock.qld.gov.au/silo/). The characteristics of the five sites are listed in Table 2.

2.3. Cultivar parameters, outputs, and management settings

Parameters can be classified into three categories: environmental parameters such as soil properties and CO_2 concentration; crop cultivar parameters; and management parameters such as fertilization and irrigation rates. The parameters belonging to the first two categories are used to initialize the simulation, while the management parameters are used to modify the environmental conditions during a specific time period. For example, the sowing date can be decided by the value of soil moisture or accumulated rainfall to make sure there is enough soil water for emergence.

Ten parameters are commonly used by APSIM-wheat to define the properties of a cultivar. New cultivars can be created by modifying the value of these parameters. The definitions, units, base values, and lower and upper bounds for the 10 parameters specified for sensitivity analysis in this study are reported in Table 3. For the bounds, we used uniform parameter distributions with the parameter ranges calculated by varying the base values by ±50% and rounded the resultant value. Then the bound of each parameter was adjusted to cover the range of values of existing cultivars. Four model outputs including yield, biomass, flowering, and maturity day were selected in this study (Table 3).

The sowing window was set from 15 May to 9 July. The sowing rules were set as: the amount accumulated rainfall in three days was larger than 30 mm and extractable soil water was larger than 200 mm. If at the end of the sowing window the sowing rules were not met, the crop was sown on the next day, 10 July. The sowing density was 250 plants m^{-2}, sowing depth was 40 mm, and row space was 250 mm. If fertilizer was applied, it was applied at the sowing day on the surface of the soil. Crop residue was left on the soil surface on harvest day.

2.4. Variance-based global sensitivity analysis method

A process-based crop model can be generalized as:

$$Y(t) = f(x, t; \theta) + \in \tag{1}$$

where $Y(t)$ is the output (state variable) for a time point t, x is the daily climate input data, θ denotes a vector of parameters, namely (P_1, P_2, . . ., P_k), and $\in$ is the residual error. In this study, we used the variance-based sensitivity analysis to study the sensitivity of the four outputs (Y) to the cultivar parameters (θ). We also investigated the impacts of management parameters and climate–soil conditions on the cultivar parameter sensitivities.

The foundation of the variance-based sensitivity analysis method is that the variance of an output (Y) of an integrable model can be decomposed accordingly into the individual parameters (P_1, P_2, . . ., P_k) and their interactions (Cukier et al., 1973; Sobol, 1990). Applying the theory to APSIM, the output Y's variance can be decomposed into the cultivar parameters:

$$V(Y) = \sum_{i=1}^{k} V_i + \sum_{i=1}^{k}\sum_{j+1}^{k} V_{ij} + \cdots + V_{1,2,\ldots,k} \tag{2}$$

where V_i denotes the variance allocated to the i-th parameter P_i. V_{ij} denotes the variance allocated to the interactions between P_i and P_j. $V(Y)$ is the total variance of Y caused by the uncertainties of all parameters. The importance of one parameter P_i to the output Y (or the sensitivity of output Y to P_i) is quantified by the ratio of P_i-caused variance to the total variance $V(Y)$, which is the main or first-order effects and is represented by the index, S_i. The equation is:

$$S_i = \frac{V_i}{V(Y)} = \frac{V[E(Y|P_i)]}{V(Y)} \tag{3}$$

where V_i denotes the variance caused by the uncertainty of the ith parameter P_i, $E(Y|P_i)$ is the conditional expectation of Y with a specific P_i value. If changing P_i value makes a significant difference for $E(Y|P_i)$, then the output Y is sensitive to P_i; otherwise P_i is non-influential. Thus, $V[E(Y|P_i)]$ can be used as a surrogate for V_i.

The total effects, ST_i accounts for all effects associated with P_i, including the main effect and the interactions with other parameters, S_{ij}. It is formalized as:

$$\begin{aligned} ST_i &= S_i + \sum\nolimits_{j \neq i} S_{ij} + \cdots \sum\nolimits_{j,k \neq i} S_{ij}\ldots, k = \frac{E[V(Y|P_{\sim i})]}{V(Y)} \\ &= 1 - \frac{V[E(Y|P_{\sim i})]}{V(Y)} \end{aligned} \tag{4}$$

$P_{\sim i}$ indicates all the other parameters except parameter P_i. For a specific output, $ST_i = 0$ implies that P_i is non-influential and can be fixed at any value without affecting the variance of the output. If $ST_i = S_i$, it means that P_i does not interact with other parameters. If $ST_i = S_i$ for all parameters, the model is additive (linear). The summation of S_i for all parameters equals 1 for a linear model. If the summation is less than 1, then the model is non-linear (Saltelli et al., 2008).

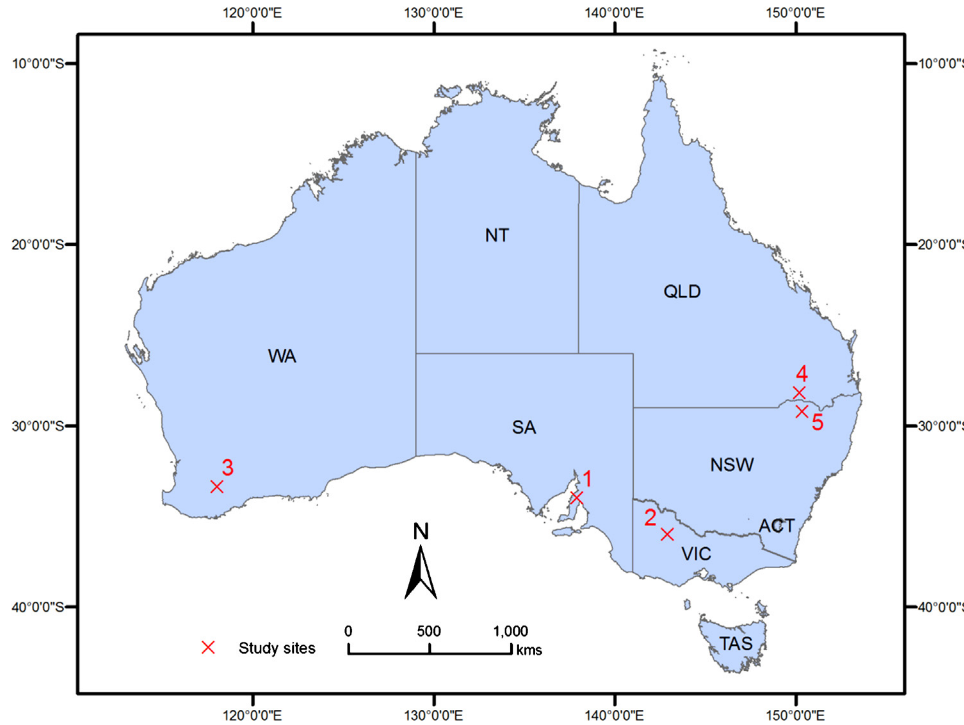

Fig. 1. The five study sites across the cereal growing regions of Australia. The detailed information of the sites can be found in Table 2. WA stands for Western Australia, SA for South Australia, NSW for New South Wales, VIC for Victoria, ACT for Australian Capital Territory, TAS for Tasmania, QLD for Queensland, and NT for Northern Territory.

Table 2
Climate and soil characteristics of the five selected sites.

	QLD	NSW	VIC	SA	WA
Site	Billa Billa	Croppa Creek	Birchip	Paskeville	Dumbleyung
Number[a]	4	5	2	1	3
Region	Darling Downs and Granite Belt	North West Slopes and Plains	Mallee	Yorke Peninsula	Central Region_Avon
Latitude	−28.162	−29.121	−35.981	−33.963	−33.341
Longitude	150.201	150.306	142.917	137.911	118.028
Mean temperature (°C)	20.19	19.58	15.71	16.41	16.29
Mean rainfall (mm yr^{-1})	569	604	333	403	339
Soil type	Red Chromosol	Grey Vertosol	Sandy Loam	Clay Loam	Sandy clay with acid
PAWC (mm)[b]	165	155	99	143	52

[a] The number is corresponding to the site number on the map of Fig. 1.
[b] Plant available water capacity down to a depth of 1 m.

Table 3
Cultivar parameters and output variables of APSIM-wheat model.

Name	Definition	Unit	Base value	Lower bound	Upper bound
Cultivar parameters					
grains_per_gram_stem	Kernel number per stem weight at the beginning of grain filling	g	25	10	40
potential_grain_filling_rate	Potential daily grain filling rate	g/grain/day	0.002	0.001	0.005
potential_grain_growth_rate	Grain growth rate from flowering to grain filling	g/grain/day	0.001	0.0005	0.0015
max_grain_size	Maximum grain size	g	0.041	0.02	0.06
tt_start_grain_fill	Thermal time from start grain filling to maturity	°Cdays	545	200	900
tt_floral_initiation	Thermal time from floral initiation to flowing	°Cdays	555	250	800
tt_flowering	Thermal time needed in anthesis phase	°Cdays	120	60	180
tt_end_of_juvenile	Thermal time needed from sowing to end of juvenile	°Cdays	400	200	600
vern_sens	Sensitivity to vernalisation[a]	–	1.5	0	5
photop_sens	Sensitivity to photoperiod[b]	–	3	0	5
Output responses					
Yield	Crop yield	kg ha^{-1}			
Biomass	Crop biomass	kg ha^{-1}			
Flowering day	Flowering day after sowing	day			
Maturity day	Maturity day after sowing	day			

[a] For winter wheat, the thermal time from the phase emergence to end of juvenile is affected by the number of cumulative vernalizing days experienced during the period. The crop development stops until the required vernalization days are attained. The default value for maximum vernalization requirement was 50 days in the model and is adjusted by the parameter of the crop's sensitivity to vernalization.
[b] The phase between end of juvenile and floral initiation is affected by a crop's photoperiod sensitivity. Standard astronomical equations were used to calculate the photoperiod with day of year and latitude as input. The relationship between photoperiod and thermal time was defined by a 2-segment linear function.

Table 4
Comparison of the four methods for computing the variance-based sensitivity indices.

	S_i	ST_i	Convergence	Run times
Monte Carlo brute force	Yes	Yes	Slow	$N^a \times K^b \times M^c \times 2$
Saltelli (2002)	Yes	Yes	Slow	$N \times (M+2)$
FAST	Yes	No	Fast	N
Extended FAST	Yes	Yes	Fast	$N \times M$

[a] N denotes the number of run times for convergence.
[b] K denotes the number of run times for computing $V[E(Y|P_i)]$.
[c] M denotes the number of parameters.

2.5. Comparison of the variance-based sensitivity indices calculation methods

A range of methods can be used to calculate the variance-based sensitivity analysis indices. In order to choose the best method for sensitivity analysis, we evaluated four methods including Monte Carlo brute-force, Saltelli's method, FAST, and extended FAST (Table 4). The Monte Carlo brute force is the most computationally demanding method (Saltelli et al., 2008). It is not feasible for relatively slow models like APSIM (APSIM needs 1 min to finish a 20-year simulation). The method proposed by Saltelli (2002) reduced the run times to $N \times (M+2)$ in calculating the main and total effects indices of all parameters, where N is the sample size for each parameter and M is the number of parameters. However, this method needs a large sample size to attain a convergence and sometimes results in negative indices (Nossent et al., 2011; Song et al., 2012). The classical FAST (Cukier et al., 1973, 1975; Schaibly and Shuler, 1973) method is fast and robust in calculating the first order indices but cannot quantify the total effects (Saltelli et al., 1999). The extended FAST method is derived from the classical FAST method and is able to calculate both the main and total effects (Saltelli et al., 1999). It computes the sensitivity indices for each parameter separately, which results in M (the number of parameters) times the computing cost of the classical FAST method. Due to the good convergence ability of the extended FAST method, small sample sizes can be used, thereby reducing the computing cost to an acceptable level. We chose this relatively computationally efficient and robust method with a sample size of 1000 for each parameter.

2.6. Effects of fertilization and climate–soil conditions

We analyzed the effects of fertilization and location on the model sensitivity by ranking the total effects of each parameter for each combination of fertilization and climate–soil conditions. There were ten combinations from 5 sites with different climate–soil conditions and two fertilization rates, 0 kg N ha^{-1} and 100 kg N ha^{-1}. The total effects were calculated for each combination by averaging the total effects of each parameter over 20 simulation years. Parameters were then ranked in the order of influence with the most influential (largest ST_i) assigned a rank of 1.

We quantified the effects of climate on the sensitivities of the ten cultivar parameters for the four model outputs by ordering the total effects of the parameters each year. For each year with its specific weather conditions, the total effects were averaged over the ten combinations of fertilization and location then parameters were ranked in order of influence.

2.7. Uncertainty analysis

We set the sample size $N = 1000$ for the sensitivity analysis to attain a stable convergence (Saltelli et al., 1999). So a total number of 10^5 ($1000 \times 10 \times 5 \times 2$) simulations were run, with 10 cultivar parameters, 5 study sites, and 2 fertilization rates. The parameter generation and indices calculation of the extended FAST was coded in the Python programming language with reference to Saltelli et al. (2000) and Song et al. (2013). For each parameter set, an APSIM .sim file was created and submitted to CSIRO's Condor grid computing system, and processed in parallel on each grid node (Zhao et al., 2013a). We collected the results for four output variables of yield, biomass, flowering day, and maturity day. We used a combination of violin and box plot to illustrate the uncertainty in outputs derived from the variation in the cultivar parameters. The values of each output for the five sites and two fertilization levels were plotted (Fig. 7).

3. Results

The grains_per_gram_stem, max_grain_size and the potential_grain_filling_rate were the most influential parameters for wheat yield, while the parameters tt_end_of_juvenile and photop_sens showed negligible effects (Fig. 2). The differences between the first order (S_i) and total effects (ST_i) for all the influential parameters were notable. Biomass was sensitive to all ten cultivar parameters, among which the vern_sens and tt_floral_initiation were the most influential ones. For biomass, the parameters tt_flowering, tt_end_of_juvenile and photop_sens had no main effects, but did have total effects. Flowering day was only sensitive to vern_sens and tt_floral_initiation. Maturity day was sensitive to tt_start_grain_fill, vern_sens and tt_floral_initiation. Both of the main and total effects sensitivity indices for flowering day and maturity day were stable with only small variations.

Fertilization had a stronger influence than climate–soil conditions on the sensitivities of yield and biomass (Figs. 3 and 4). For example, in NSW applying 100 kg N ha^{-1} changed the order of potential_grain_filling_rate from the second most influential to the sixth. In QLD, the order of tt_start_grain_fill changed from fifth to first. The impact of climate–soil conditions was stronger under a fertilization level of 100 kg N ha^{-1}. The parameter tt_start_grain_fill was the sixth most influential in SA but first in QLD. The order of least and most influential cultivar parameters for yield did not change over the 20-year simulation (grains_per_grain_stem and potential_grain_growth_rate) (Figs. 5 and 6).

Uncertainty in model outputs was significant under the specified variation in input parameters. With no fertilization, yield ranged from 64 to 3559 kg ha^{-1} (minimum and maximum), biomass from 693 to 12,864 kg ha^{-1}, flowering day from 63 to 189 days, and maturity day from 82 to 221 days (Fig. 7). Fertilization of 100 kg N ha^{-1} significantly increased the variation ranges (maximum yield 9157 kg ha^{-1}, biomass 22,057 kg ha^{-1}) and extended the distributions for yield and biomass, except for QLD where the fertilization decreased the yield distribution. Fertilization did not influence the ranges of the flowering day and maturity day for NSW, VIC, and WA, but delayed the flowering day and maturity day in QLD and WA.

4. Discussion

Using the extended FAST variance-based sensitivity analysis method, we investigated the sensitivity of four APSIM outputs—yield, biomass, flowering day and maturity day—to variations in ten cultivar parameters. We explored the effects of fertilization rates and climate–soil conditions on sensitivity indices across five sites in the Australian croplands. We also quantified the uncertainty derived from the variation of the cultivar parameters for the four outputs.

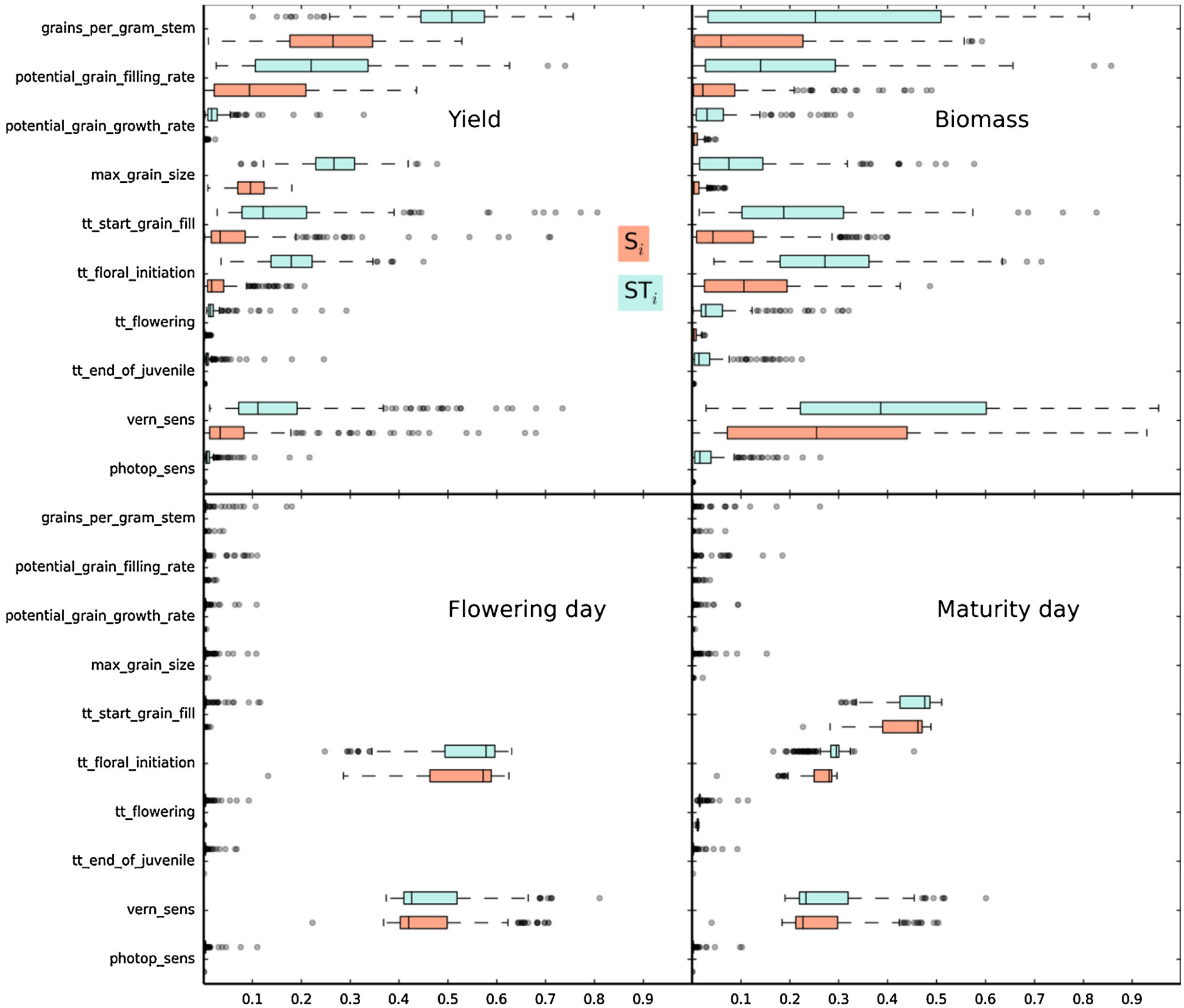

Fig. 2. Variations of the total (ST_i) and main (S_i) effect indices for the four outputs to ten cultivar parameters in APSIM-wheat under two fertilization rates, over 20-year, across five locations. The line in the box is the median, the edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to 1.5 × (Q3 − Q1) beyond which was indicated by the dots, fliers.

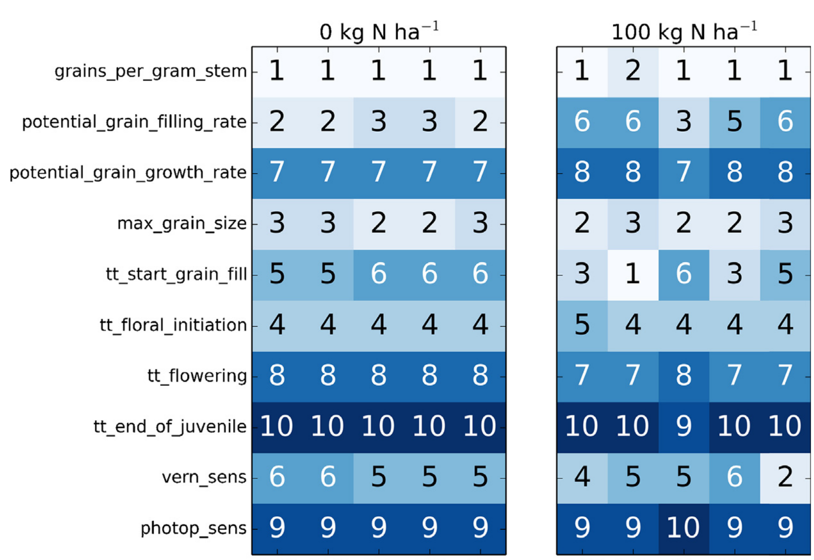

Fig. 3. Impact of management and climate–soil conditions on the order of yield sensitivity (total effects) to ten cultivar parameters.

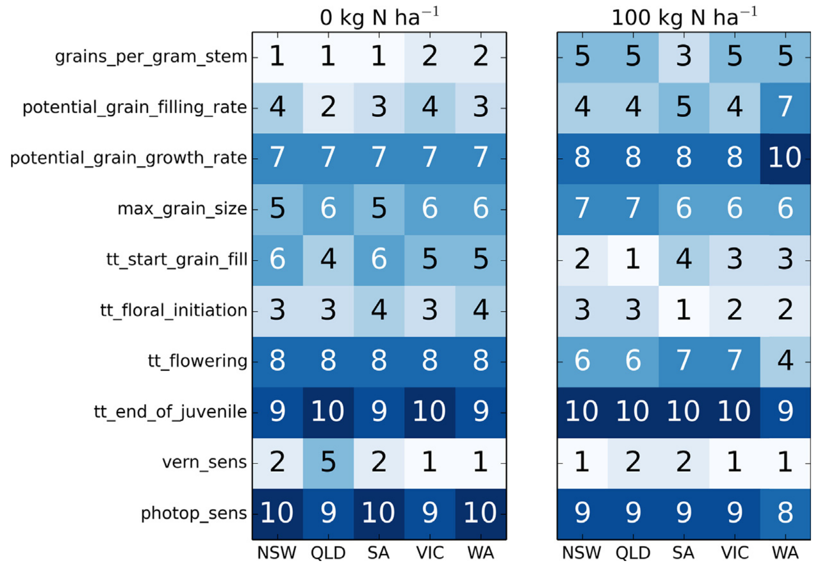

Fig. 4. Impact of management and climate–soil conditions on the order of biomass sensitivity (total effects) to the ten cultivar parameters.

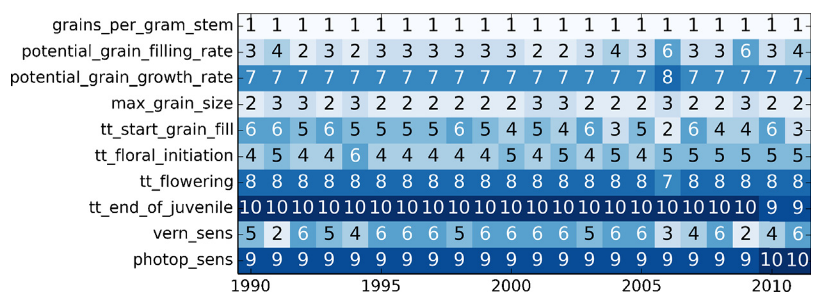

Fig. 5. Time dependent order of yield sensitivity (total effects) to the ten cultivar parameters—impacts from climate.

4.1. Sensitivity of outputs to cultivar parameters

Wheat yields were mostly sensitive to three parameters: grains_per_gram_stem, max_grain_size and potential_grain_filling_rate. The first two are the essential parameters that determine the yield component of wheat and the third parameter adjusts the grain filling rate in the reproductive stage (e.g. time between the flowering day and maturity day). In the APSIM-wheat module, yield is calculated as the product of grain number and grain size. The grain number is calculated at the flowering stage as a function of grains_per_gram_stem and the dry matter accumulated between the stages of flag leaf stage and the start of grain filling. The wheat grain size is computed by grain growth rate (potential_grain_filling_rate) and the partitions of biomass to the grain in the reproductive stage after flowering (Heiniger et al., 1997). The other influential parameters for yield such as tt_start_grain_fill, vern_sens and tt_floral_initiation were correlated with the phenology. This was shown in the second row of the box plots (Fig. 2). The length of the crop growth, especially at the reproductive stage, affected how much the biomass can be accumulated and then allocated to the wheat grain.

Biomass was strongly influenced by the vern_sens and tt_floral_initiation. In APSIM-wheat, the leaf (indicated by LAI) and dry matter biomass grow quickly in the stages after emergence and before the flowering day. The thermal time needed between the stages from emergence to end of juvenile was determined by the vernalization sensitivity of the cultivar and the number of vernalization days during the period (Zhang et al., 2012). Changing the value of vern_sens resulted in a prolonged or shortened vegetative (LAI, dry matter) growth period, thereby affecting the biomass accumulation. APSIM used the parameter of tt_floral_initiation to specify the thermal time needed from

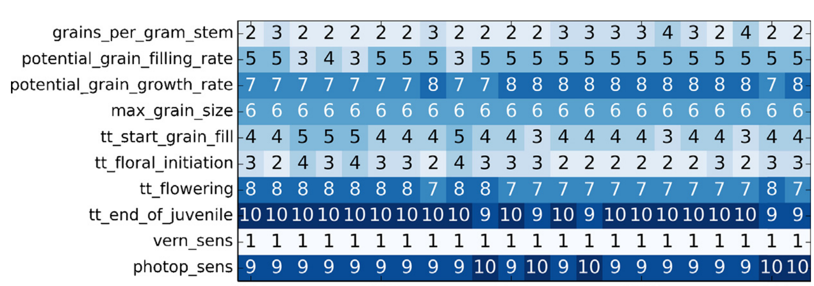

Fig. 6. Time dependent order of biomass sensitivity (total effects) to the cultivar parameters—impacts from climate.

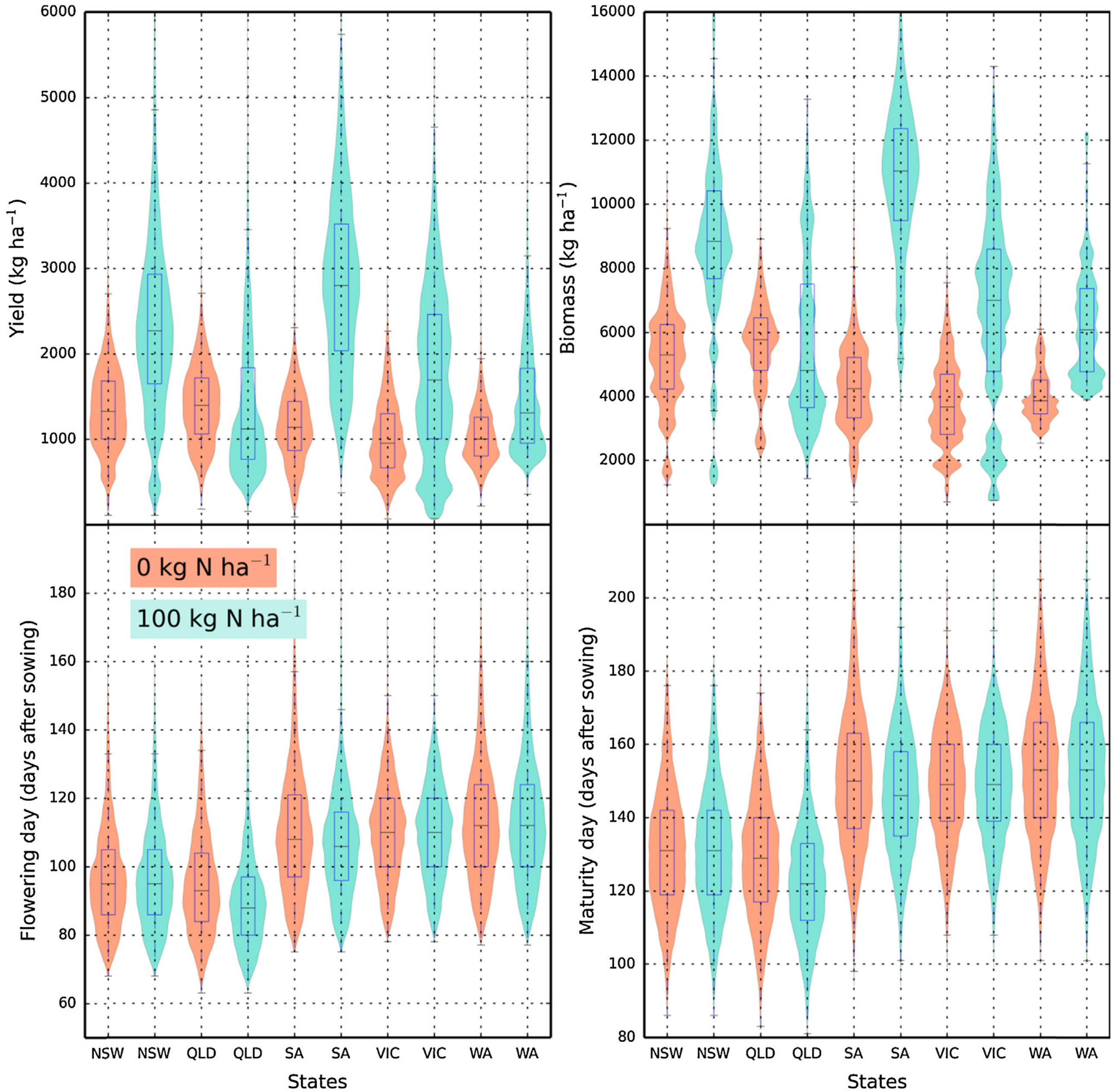

Fig. 7. The empirical frequency distributions for APSIM-wheat output variables under two levels of fertilization rates across five sites in Australia crop lands. The violin plot and the box plot were based on the same data. The line in the box is the median, the edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to 1.5 × (Q3 − Q1).

the floral initiation to flowering stage. As the grains were part of the total biomass, the influential parameters for yield were also important to the biomass accumulation. The biomass sensitivity indices varied significantly, which illustrated that biomass sensitivity to the cultivar parameters was strongly influenced by the environmental conditions and fertilization rates.

Both of the flowering day and maturity day were sensitive to vern_sens and tt_floral_initiaition, while the tt_start_grain_fill affected only the maturity day. The tt_start_grain_fill determined the length of the start grain filling stage which was between flowering and maturity, so it was important to the maturity day but not flowering day. These three important parameters should be included when calibrating the phenology of APSIM with the best fit to observed records. While parameters that determine yield components will have no effect on crop phonological development, parameters that determine development will strongly influence yield, and its components, though effects on the duration of phases for canopy development and biomass accumulation. Thus, in the model calibration process, the phenology parameters should be calibrated before the parameters related to the yield component (Ma et al., 2011).

The differences between the total effects (ST_i) and the first order effects (S_i) were large for yield and biomass, but very small for flowering day and maturity day. This suggests that there were non-linear relationships involved in simulating yield and biomass, with less non-linearity in simulating the flowering day and maturity day. Non-linear relationships resulted from the interactions between the influential cultivar parameters, climate–soil conditions, and management practices. The quantification of the interactions among these factors provided insight into model behaviour, which cannot be detected by local sensitivity analysis method or the classic FAST method.

All of the four outputs were not sensitive to photop_sens (sensitivity to photoperiod), which did not match our expectation. The APSIM documentation stated that APSIM-wheat sets the end of the juvenile stage as the second day after emergence, because the growth rate of wheat was sensitive to photoperiod from emergence rather than the end of juvenile stage (Keating and Meinke, 1998; Wang et al., 2002). By inspecting the code, we found that in APSIM the photop_sens is converted to photoperiod effects by multiplying a factor of 0.002 (according to the source code). This could be the reason that the influence from photop_sens was so small. This suggests that the photop_sens should be set as a relatively large value for crops with growth affected by day length. The non-influential result for photop_sens may result from the narrow range of the parameter (0–5).

4.2. Effects of fertilization and climate–soil conditions

The effects of fertilization were stronger than climate–soil condition on the order of the parameter sensitivities for yield and biomass (Figs. 3 and 4). This indicated that the interactions of cultivar parameters with fertilization rates were stronger than with climate–soil conditions (Dzotsi et al., 2013). Biomass and yield were mainly limited by radiation, water, and nutrients. Crop growth would be stressed by a lack of nitrogen availability. Consequently, the photosynthesis, expansion, phenology and tillering would all be affected. Application of 100 kg N ha^{-1} fertilizer transferred the limiting factor from nitrogen to water or radiation. So in this scenario, the influence of climate–soil condition became stronger with 100 kg N ha^{-1} fertilizer application. The most common N application rates in Australia are between 50 and 80 kg ha^{-1} (ABARE-BRS, 2003). Due to computational constraints, we chose to analyze sensitivity at two extreme fertilization rates rather than the most common. The effects of fertilization could be further explored by analysing more fertilization rates.

The temporal characteristics of the parameter sensitivity showed that the rank of parameters only changed slightly over the twenty-year simulation period. The temporal simulation used the same soil parameter for each site and impacts mainly resulted from climatic variation, which did not vary dramatically across different years, thus the impacts were minor (Pogson et al., 2012).

4.3. Uncertainty of the outputs resulting from cultivar parameters

Uncertainty in the model outputs resulting from variation in cultivar parameters was large. Hence, the values of the cultivar parameters, especially the most influential ones, should be carefully determined, otherwise unreliable model results are likely (DeJonge et al., 2012). Cultivar parameters are normally determined by literature review, expert opinion or calibration against trial data. As certain parameters could vary from one environmental condition to another, the uncertainties derived from the cultivar parameters should always be considered when making decisions based on simulated results. In order to reach a desired level of confidence in the results and offer robust information for decision making, the uncertainty for the most influential parameters should be preferentially reduced by pertinent calibration and validation (Aggarwal, 1995). Fertilization had some effects at the sites located in QLD and SA. This may be due to the soil water resetting issue. Applying 100 kg N ha^{-1}, more soil water would be consumed in the growing season due to higher biomass production. This left the soil drier and may influence the sowing decision in following year (e.g. fertilization may delay sowing in the following year). A later sowing then could cause later flowering days and maturity days.

4.4. Sensitivity calculation method

We chose the extended FAST variance-based sensitivity analysis method over other methods for calculating sensitivity indices. The advantage of the extended FAST method is its fast convergence with an acceptable computing cost (Saltelli et al., 1999; Song et al., 2013; Wang et al., 2013a). APSIM is a relatively computationally intensive model, so a sensitivity analysis method that needs the least model runs is desirable (Wallach et al., 2006). The computation cost of the extended FAST method is M (number of parameters) times the classic FAST, but it enables calculation of the total effects. Thus, the important interactions between parameters can be investigated (Wang et al., 2013b). The computing challenge of the sensitivity was overcome by the grid computing and parallel programming techniques. Since there was no interaction between model runs, parallelism could be applied to the simulations (Zhao et al., 2013a). Our experience with the computation cost for the different indices calculation methods could help other practitioners make a wise choice in application of the global sensitivity analysis methods to different crop models.

4.5. Limitations

Firstly, the uncertainty ranges of the cultivar parameters are unavailable neither in the APSIM documentation nor literature. Instead, we set the ranges as ±50% of the default base cultivar values. If the cultivar parameter is influential, narrowing or broadening the range of the parameter could influence the parameter sensitivity significantly (Wang et al., 2013b). Refining the uncertainty ranges of the parameter, especially the most influential ones could further improve the sensitivity results. Secondly, we only investigated the interactions between cultivar, nitrogen management and climate–soil conditions. APSIM comprises many other types of parameter. These parameters could also have interactions with the cultivar parameters and could influence their importance to the selected outputs. Classifying these parameters to different groups and assessing the relative importance of these groups with the variance-based method could further clarify the model's structure (Song et al., 2012).

5. Conclusion

Sensitivity of four APSIM outputs—yield, biomass, flowering day, and maturity day—to ten cultivar parameters of wheat were analyzed using a variance-based global sensitivity analysis method. The yield was mostly affected by the cultivar parameters that determine yield components (grains_per_gram_stem, max_grain_size, and potential_grain_filling_rate) and the length of the key reproductive stages (tt_floral_initiation and tt_start_grain_fill). All the cultivar parameters had total effects on biomass. Among them vern_sens (vernalization sensitivity) and tt_floral_initiation were the most influential ones. Both the phenological outputs of flowering day and maturity day were sensitive to vern_sens and tt_floral_initiation. The effects of fertilization were stronger than climate–soil condition on the order of the parameters' impacts on yield and biomass. Fertilization significantly increased the variation in yield and biomass, but had negligible effects on flowering day and maturity day. We conclude that by applying a strategy to calibrate the influential phenological parameters firstly and then the yield component parameters could speed up the calibration of APSIM. The comprehensive sensitivity results from this study could serve as a valuable complement for the documentation of APSIM and inform similar sensitivity analyses of other crop models.

This study showed that global sensitivity analysis can be employed to study the importance of parameter to various outputs in a process-based crop model. The results revealed the subset of parameters that explains the most of the variance for different outputs. For a specific output, parameters with a small impact can be excluded in the calibration exercise. This could be useful for calibration of process-based crop models for multiple sites or at a regional scale (Minunno et al., 2013).

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